

Disharee Bhowmick

Samuel Ginn College of Engineering | Auburn University, Alabama USA, 36849
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Education

Auburn University, Alabama

Ph.D. in Computer Science

Jan 2024 – present

GPA: 3.83

Auburn University, Alabama

Master of Science in Computer Science

Aug 2023 – May 2025

GPA: 3.83

University of Engineering and Management, Kolkata

Bachelor of Technology in Computer Science

Aug 2015 – Jul 2019

GPA: 3.6

Teaching Experience (GTA, Auburn University)

- Lab Instructor, Introduction to Engineering (ENGR 1110)
- Lead Teaching Assistant to Dr. Vishalini Ramnath, Introduction to Algorithms (COMP 3270)
- Teaching Assistant to Dr. Vishalini Ramnath, Formal Languages (COMP 4200)

Research Experience (GRA, Auburn University)

Graduate Research Assistant

CSSE Department, Auburn University | Advisor: Dr. Sathyanarayanan N. Aakur

Summer 2024 – present

Summer 2024 & Summer 2025 — Time Series Summarization (STaTS)

- Developed STaTS, a lightweight unsupervised framework for structure-aware temporal sequence summarization, adaptively compressing time series using a BIC-based multi-scale change detection criterion.
- Achieved up to 30x input compression while retaining 85-90% of full-resolution model accuracy across 150+ datasets spanning UCR, UEA, and ETT benchmarks.
- Work published in **Pattern Recognition Letters**, Vol. 202, pp. 135-140, 2026.

Summer 2026 — Adapting VLMs for Open World Event Segmentation

- Developing and evaluating temporal segmentation pipelines that adapt Vision-Language Models (VLMs) for open-world event segmentation, enabling recognition and localization of action categories beyond closed-set training distributions.

Publications

- Disharee Bhowmick, Ranjith Ramanathan, Sathyanarayanan N. Aakur. STaTS: Structure-aware temporal sequence summarization via statistical window merging. *Pattern Recognition Letters*, Vol. 202, pp. 135-140, 2026.
- Revanth Reddy Palem, Mariana E. Garcia-Ascolani, Disharee Bhowmick, David Lalman, Ranjith Ramanathan, Sathyanarayanan N. Aakur. Vision-Based Livestock Respiratory Monitoring for Smart Agriculture Sensing Systems. *IEEE Midwest Symposium on Circuits and Systems (MWSCAS)*, 2026.

Latest Preprints

- Shanmukha Vellamcheti, Uday Kiran Kothapalli, Disharee Bhowmick, Sathyanarayanan N. Aakur. CVT-Bench: Counterfactual Viewpoint Transformations Reveal Unstable Spatial Representations in Multimodal LLMs. *arXiv:2603.21114*, 2026.
- Revanth Reddy Palem, Disharee Bhowmick, Mariana E. Garcia-Ascolani, David L. Lalman, Ranjith Ramanathan, Sathyanarayanan N. Aakur. Physiological Signal-Aware Livestock Respiration Estimation from Video.

Employment History

Student Assistant — Innovation and Research Commons (I&RC)

Ralph Brown Draughon Library, Auburn University

Summer 2025

- Assisted patrons with multimedia, MakerSpace, and Adobe Creative Cloud resources, including 3D printing, VR/AR equipment, and digital media production tools.

Technician — Jay and Susie Gogue Performing Arts Center

Fall 2023

Auburn University

- Assisted with technical operations supporting live performances and events at the venue.

Cognizant Technology Solutions

Sep 2019 – Jun 2023

Software Engineer → Data Analyst → ML Engineer | Pune, Maharashtra, India

Phase 1 — Foundation & Training (6 months, Sep 2019 – Feb 2020)

- Completed structured onboarding covering data analysis fundamentals, exploratory data analysis (EDA), statistical methods, and data preprocessing pipelines.

Phase 2 — Data Analyst (1 year, Mar 2020 – Feb 2021)

- Developed Python-based analytics dashboards using Power BI and Tableau to monitor IT incident trends for enterprise clients.
- Partnered with operations teams to identify root causes of recurring incidents; implemented data-driven interventions that **reduced incident volume by 25%**.

Phase 3 — Machine Learning Engineer, Internal Project (1 year, Mar 2021 – Feb 2022)

- Led an internal ML project focused on predictive analytics; designed and evaluated supervised learning models using Python (scikit-learn, pandas).
- Performed feature engineering, model validation, and performance benchmarking across multiple ML architectures.

Phase 4 — Generative AI (16 months, Mar 2022 – Jun 2023)

- Built and deployed Generative AI solutions integrating large language models into enterprise workflows.
- Designed and implemented **Retrieval-Augmented Generation (RAG)** pipelines to ground LLM outputs in enterprise knowledge bases, improving response accuracy and reducing hallucinations.
- Performed **fine-tuning** of large language models on domain-specific datasets to adapt pre-trained models for enterprise use cases, improving task performance significantly.

Intern — Ogma Tech Lab

Jul 2018 – Aug 2018

Kolkata, India

- Developed an Android Studio-based "Puja Parikrama" application using Android Studio and Firebase.

Skills & Courses

- Mathematics & Statistics: Basic statistics, probability theory, statistical tests, time series concepts (autocorrelation, seasonality, stationarity)
- Programming: Python, NumPy, pandas, scikit-learn, TensorFlow, PyTorch
- Data & ML: Cleaning, missing data handling, feature engineering, supervised/unsupervised learning, model evaluation, ensembles
- ML Frameworks: RNNs, LSTMs, GRUs, CNNs, Attention, Transformers
- Data Visualization: Power BI, Tableau
- GenAI & CV: Large Language Models, vision-based generative AI, computer vision pipelines
- Courses: Deep Learning, Computational Imaging, Machine Learning, Image & Video Processing, Natural Language Processing, Multimedia & Web Databases

Honors and Awards

- Recipient of the 100+ Women Strong Outstanding Departmental Annual Graduate Award (2026), Samuel Ginn College of Engineering, Auburn University — selected as one of two graduate student honorees from the Department of Computer Science and Software Engineering. Award value: \$2,000.
- Only CSSE nominee for the ACM SIGHPC Fellowship by Auburn University.
- Received Rising Star Award for project work from Cognizant Technology Solutions.
- Awarded Star of the Month multiple times at Cognizant.